

Redox Reaction

Oxidation:

- 1- Oxygen is gained
- 2- Electrons are lost
- 3- Rise in oxidation number

Reduction:

- 1- Oxygen is lost
- 2- Electrons are gained
- 3- Fall in oxidation number

Oxidation and reduction always take place together, the reaction is called redox reaction

Properties when assigning oxidation numbers:

1- Fluorine: 1-

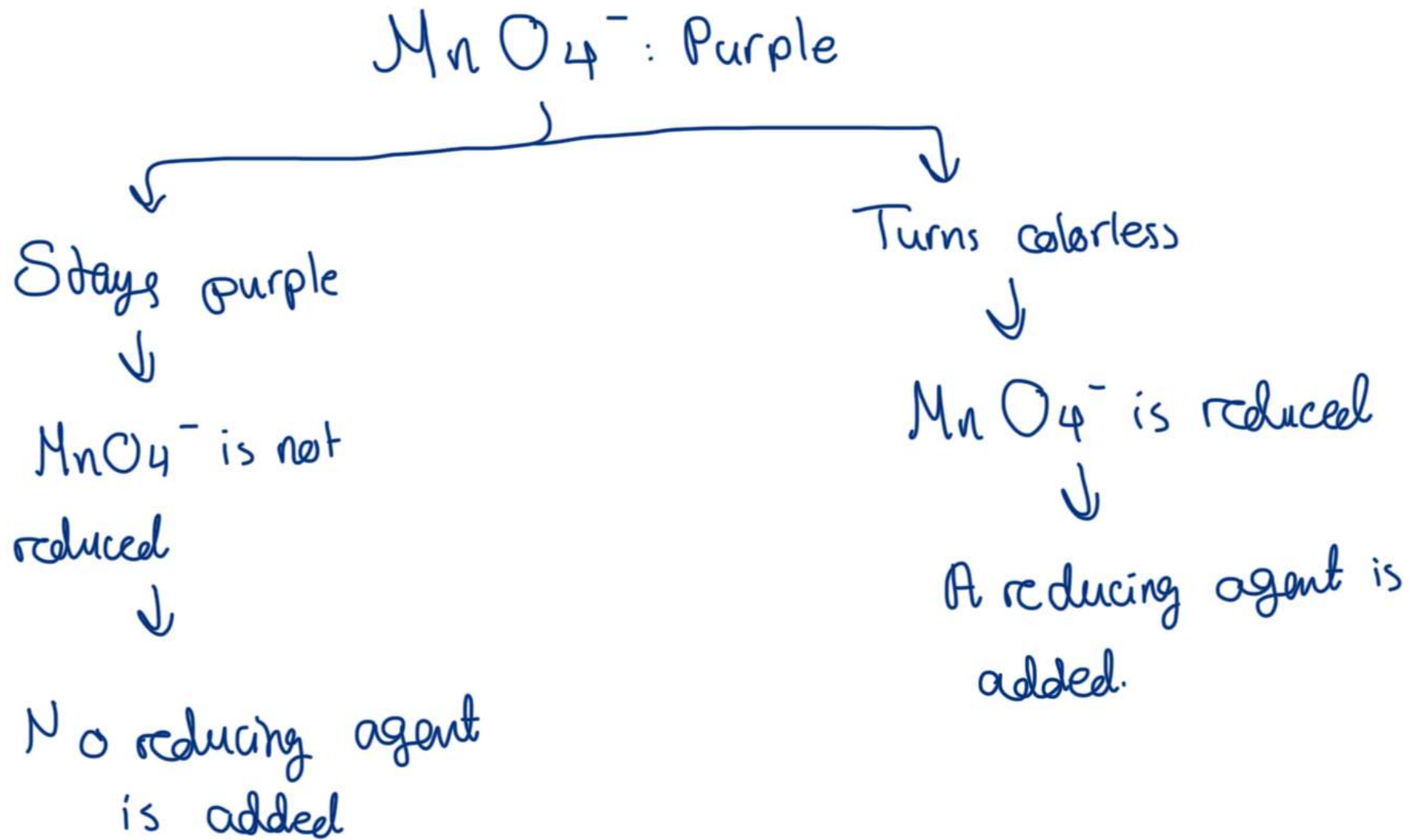
2- Oxygen can be -2, or -1 when in peroxides and 1+ in Fluorine.

3- Halogens: Without oxygen: -1, with oxygen + #

4- Hydrogen: With a non-metal: +1, with a metal: -1

Potassium manganate (VII): Purple

\* Oxidizing agent  $\rightarrow$  to be reduced



Potassium iodide: Colorless

\* A reducing agent  $\rightarrow$  to be oxidized

$I^-$ : Colorless

